

HDOC 18

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Batch: 16

1. Algorithm

1. Start main program.
2. Display the menu with 4 options to the user.
3. Read the user's choice.
4. Use a switch statement to call: `checkStrong()`, `checkArmstrong()`, `checkHarshad()`, or `checkDuck()`.
5. Function **checkStrong()**:
 - Read positive integer num. Initialize `sum = 0` and `temp = num`.
 - Loop while `temp > 0`: Extract `rem = temp % 10`. Calculate factorial of rem. Add factorial to sum. Update `temp = temp / 10`.
 - If `sum == num`, it's a Strong number. Else, not a Strong number.
6. Function **checkArmstrong()**:
 - Read positive integer num. Count the number of digits in num and store in `digits`.
 - Initialize `sum = 0` and `temp = num`.
 - Loop while `temp > 0`: Extract `rem = temp % 10`. Add `rem^digits` to sum. Update `temp = temp / 10`.
 - If `sum == num`, it's an Armstrong number. Else, not an Armstrong number.
7. Function **checkHarshad()**:
 - Read positive integer num. Initialize `sum = 0` and `temp = num`.
 - Loop while `temp > 0`: Extract `rem = temp % 10`. Add rem to sum. Update `temp = temp / 10`.

- If $\text{num} \% \text{sum} == 0$, it's a Harshad number. Else, not a Harshad number.

8. Function **checkDuck()**:

- Read positive integer num. Initialize $\text{isDuck} = 0$ and $\text{temp} = \text{num}$.
- Loop while $\text{temp} > 0$: Extract $\text{rem} = \text{temp} \% 10$. If $\text{rem} == 0$, set $\text{isDuck} = 1$ and break. Update $\text{temp} = \text{temp} / 10$.
- If $\text{isDuck} == 1$, it's a Duck number. Else, not a Duck number.

9. Stop

2. Test Case Table

TC ID	Scenario Description	Inputs (choice, num)	Expected Output
TC-01	Strong number check	Choice: 1 Num: 145	Output: Strong Number
TC-02	Armstrong number check	Choice: 2 Num: 153	Output: Armstrong Number
TC-03	Harshad number check	Choice: 3 Num: 18	Output: Harshad Number
TC-04	Duck number check	Choice: 4 Num: 102	Output: Duck Number
TC-05	Negative test case (Not Harshad)	Choice: 3 Num: 19	Output: Not a Harshad Number
TC-06	Invalid menu choice	Choice: 5	Invalid choice.

3. C Program Source Code

```
#include <stdio.h> #include
<math.h>

// Function Prototypes (void return type, no arguments) void checkStrong(void);
void checkArmstrong(void); void
checkHarshad(void); void
checkDuck(void);

int main() {
    int choice;
    printf("--- HDOC 18 Menu ---\n");
    printf("1. Check whether it is a Strong number\n"); printf("2. Check whether it is an
    Armstrong number\n"); printf("3. Check whether it is a Harshad number\n"); printf("4.
    Check whether it is a Duck number\n"); printf("Enter your choice (1-4): ");
    scanf("%d", &choice);

    switch (choice) {
        case 1: checkStrong(); break; case 2:
        checkArmstrong(); break; case 3:
        checkHarshad(); break; case 4: checkDuck();
        break;
        default: printf("Invalid choice.\n"); break;
    }
    return 0;
}

// Function 1: Check Strong Number void
checkStrong(void) {
    long long num, temp;
    int rem, sum = 0, fact, i;

    printf("Enter a positive integer: "); scanf("%lld", &num);
    if (num <= 0) { printf("Error: Positive integer required.\n"); return; }

    temp = num;
    while (temp > 0) { rem = temp
        % 10; fact = 1;
        for (i = 1; i <= rem; i++) { fact *= i;
        }
        sum += fact; temp /=
        10;
    }
}
```

```

    if (sum == num) printf("Number: %lld, Output: Strong Number\n", num); else printf("Number: %lld,
    Output: Not a Strong Number\n", num);
}

```

// Function 2: Check Armstrong Number void

```

checkArmstrong(void) {
    long long num, temp, sum = 0; int rem, digits =
    0;

    printf("Enter a positive integer: "); scanf("%lld", &num);
    if (num <= 0) { printf("Error: Positive integer required.\n"); return; }

    temp = num;
    while (temp > 0) { digits++; temp /= 10; } // Count digits

    temp = num;
    while (temp > 0) { rem = temp
        % 10;
        sum += round(pow(rem, digits)); // Using round for precision temp /= 10;
    }
    if (sum == num) printf("Number: %lld, Output: Armstrong Number\n", num); else printf("Number: %lld,
    Output: Not an Armstrong Number\n", num);
}

```

// Function 3: Check Harshad Number void

```

checkHarshad(void) {
    long long num, temp; int rem,
    sum = 0;

    printf("Enter a positive integer: "); scanf("%lld", &num);
    if (num <= 0) { printf("Error: Positive integer required.\n"); return; }

    temp = num;
    while (temp > 0) { rem = temp
        % 10; sum += rem;
        temp /= 10;
    }
    if (num % sum == 0) printf("Number: %lld, Output: Harshad Number\n", num); else printf("Number: %lld, Output:
    Not a Harshad Number\n", num);
}

```

// Function 4: Check Duck Number void

```

checkDuck(void) {
    long long num, temp; int rem,
    isDuck = 0;

    printf("Enter a positive integer: "); scanf("%lld", &num);

```

```
if (num <= 0) { printf("Error: Positive integer required.\n"); return; }

temp = num;
while (temp > 0) {
    rem = temp % 10;
    if (rem == 0) {
        isDuck = 1;
        break;
    }
    temp /= 10;
}
if (isDuck) printf("Number: %lld, Output: Duck Number\n", num);
else printf("Number: %lld, Output: Not a Duck Number\n", num);
}
```